Business Statistics

Shape of the distribution



Some graphics and datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Measures of Dispersion Overview

- What is the “dispersion” of a variable?
  - Spread and shape
  - How the value is “dispersed” or spread out around the “center”
- Value-based dispersions
  - Mean Absolute Deviation (MAD)
  - Mean Absolute Percentage Deviation (MAPD)
  - Variance
  - Standard Deviation
  - Coefficient of Variation
  - Skew
  - Kurtosis

Dispersion of a Variable
What is the shape of the distributed values?

- The shape of a variable's distribution depends on how many observations are located where
- Relative to the mean, it may look relatively:
  - Symmetric with the same shape on both sides of the mean
  - Asymmetric with more values, or a different layout of observations on one side of the mean than on the other
  - Peaked with many values in narrow area or in a few bins of a histogram
  - Tails with few values that extend far from the mean
  - Combinations of all the above

Shape of the distribution

Skew and kurtosis



The shape of a variable
What is the shape of the distributed values?

- Mean:
  - Location of the “center” of the variable’s observations
- Variance:
  - Spread of the observations about the mean (center)
- Skew:
  - Symmetry (lack thereof) of the observations on either side of the mean
- Kurtosis:
  - “Heaviness” of the tails relative to the normal distribution

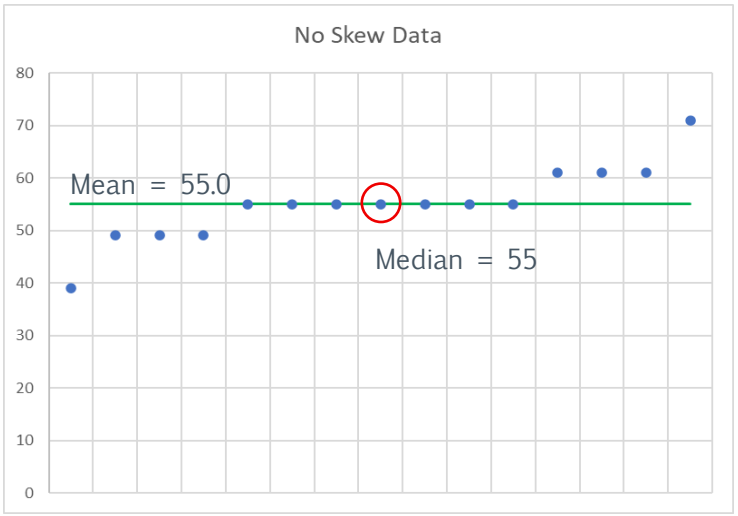
The shape of a variable
Skew is symmetry or not about the mean

- No skew: symmetric about the mean
  - mean = median skew = 0
- Positive skew: tail extends to the right
  - mean > median skew > 0
- Negative skew: tail extends to the left:
  - mean < median skew < 0

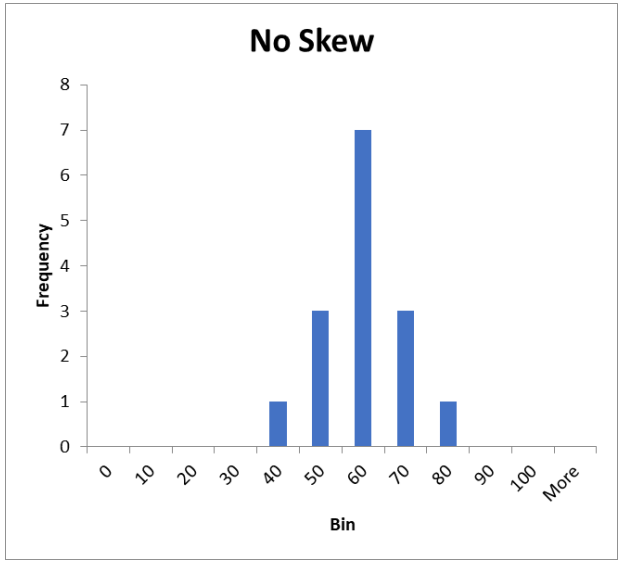
The shape of a variable
Symmetrical, no skew

- Symmetrical distribution of observations around the mean
- Values are distributed the same on both sides of the mean
- Each side is a mirror image of the other about line drawn at the mean (bin)
- mean = median
- skew = 0

39	49	49	49	55	55	55	55	55	55	55	61	61	61	71



Bin	Frequency
0	0
10	0
20	0
30	0
40	1
50	3
60	7
70	3
80	1
90	0
100	0
More	0

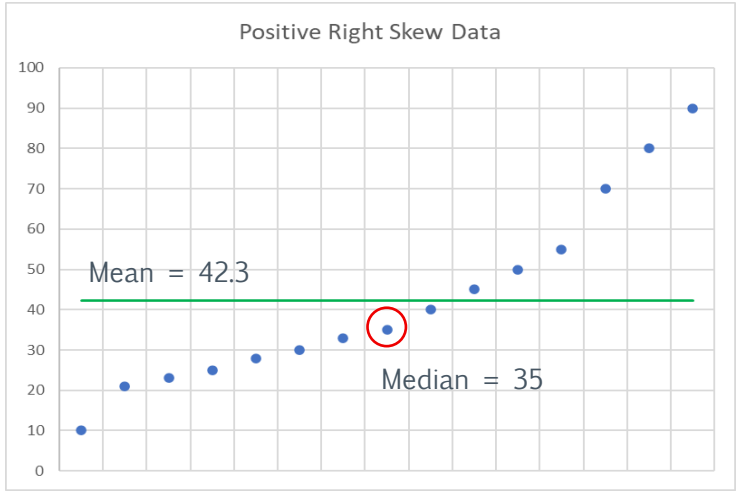


Statistics	No Skew
n	15
Median	55
Mean	55.0
Variance	52.00
Std. Dev.	7.2
CV	0.13
Skew	0.00

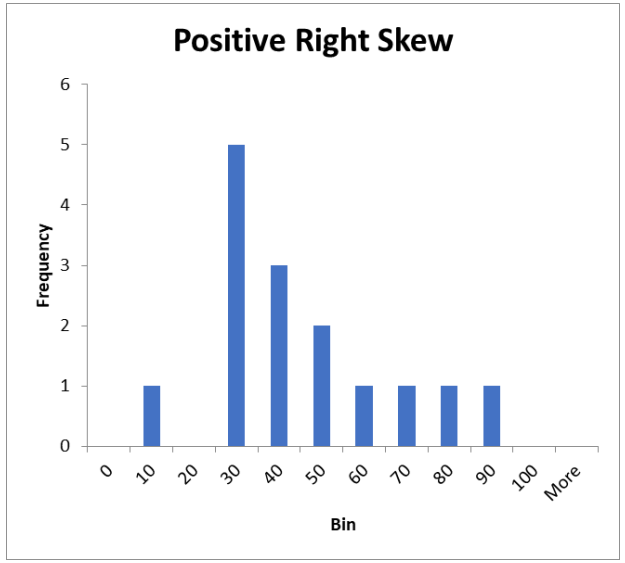
The shape of a variable
Positive skew with tail to the right

- Majority of observations are below (to the left) of the mean
- Fewer observations to the right, but they are further out
- These “pull” the tail to the right
- mean > median
- skew > 0

10	21	23	25	28	30	33	35	40	45	50	55	70	80	90



Bin	Frequency
0	0
10	1
20	0
30	5
40	3
50	2
60	1
70	1
80	1
90	1
100	0
More	0

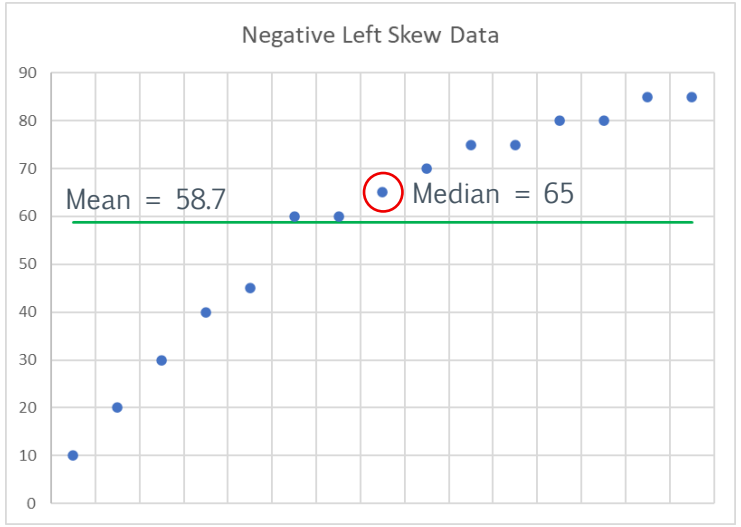


Statistics	Right (+) Skew
n	15
Median	35
Mean	42.3
Variance	525.81
Std. Dev.	22.9
CV	0.54
Skew	0.84

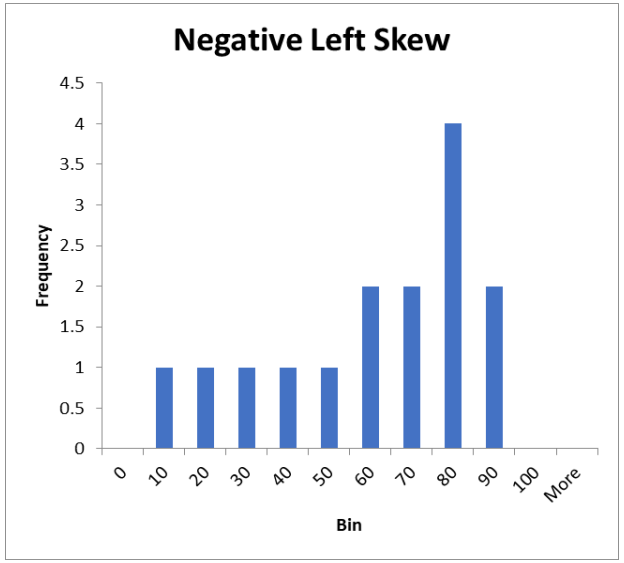
The shape of a variable
Negative skew with tail to the left

- Majority of observations are above (to the right) of the mean
- Fewer observations to the left, but they are further out
- These “pull” the tail to the left
- mean < median
- skew < 0

10	20	30	40	45	60	60	65	70	75	75	80	80	85	85



Bin	Frequency
0	0
10	1
20	1
30	1
40	1
50	1
60	2
70	2
80	4
90	2
100	0
More	0



Statistics	Left (-) Skew
n	15
Median	65
Mean	58.7
Variance	587.38
Std. Dev.	24.2
CV	0.41
Skew	-0.81

The shape of a variable
Kurtosis is about outliers in the distribution

- A higher kurtosis indicates “heavier” tails which have outliers
- A “heavy” tail has a higher chance of the variable taking on values far from the mean, including outliers
- A distribution’s kurtosis is compared to that of the normal distribution
- Kurtosis = 3 for normal distribution

- Leptokurtic (high kurtosis):
  - Kurtosis > 3
  - Heavier tails, more outliers, sharper peak
- Platykurtic (low kurtosis):
  - Kurtosis < 3
  - Lighter tails, fewer outliers, flatter peak

